

Extending surface bio-optical properties to depth: a neural network for merging ocean color and Argo data

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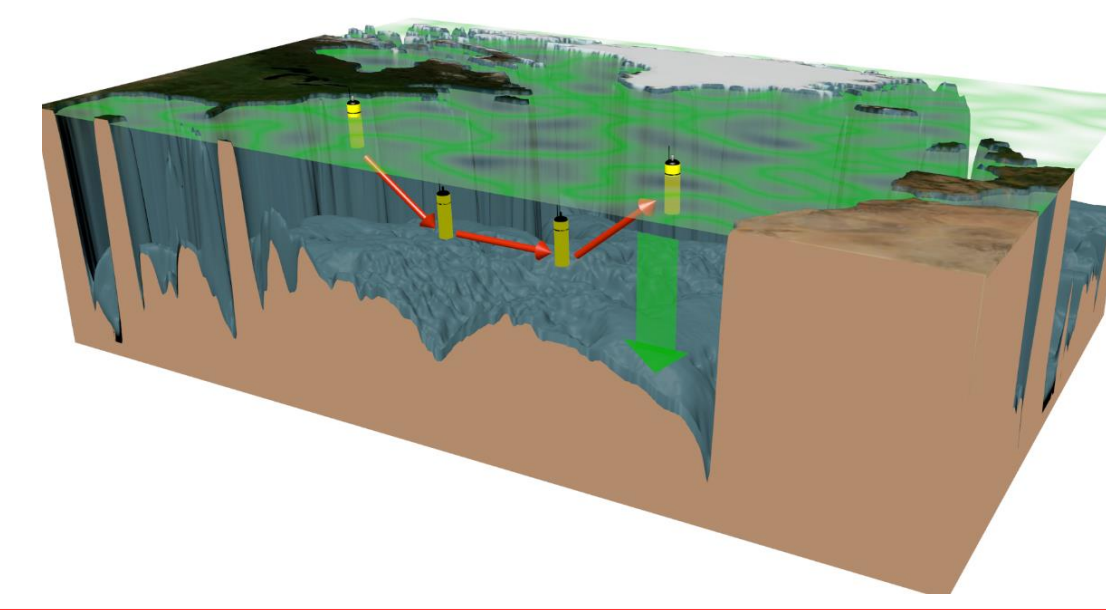
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Aim of the study

Coupling satellite **ocean color** observations with **Argo profiles** (T/S only) to infer the vertical distribution of 1) the optical particulate **backscattering** coefficient (b_{bp}) and 2) the **chlorophyll concentration** ([Chl]) associated with the total phytoplankton biomass and three size classes (micro-, nano-and pico-phytoplankton)

Argo T/S profile and ocean color matchup → Vertical profiles of b_{bp} and [Chl]



Methodology

Using a **Multi-Layered Perceptron (MLP) neural network**, the method **SOCA** (*Satellite Ocean Color and Argo data to vertical distribution of bio-optical properties*) **retrieves the vertical distribution of remotely sensed bio-optical properties in the global ocean**. 2 MLPs (referred to as SOCA-BBP for b_{bp} , and SOCA-CHL for [Chl]) were **trained and validated** using concurrent profiles of density and bio-optical properties especially collected by **Bio-Argo profiling floats**.

Ocean color data:

- MODIS-Aqua (9km / 8day)
- For b_{bp} : $\log(b_{bp}(700nm))$ from QAA (Lee et al., 2002) and $\log([Chl])$
- For [Chl]: $\log([Chl])$ and PAR

Properties from Argo T/S vertical profiles:

- 4 density values
- Z_m (mixed layer depth), from de Boyer Montégut et al., (2004)
- Z_{norm} (normalization depth defined as depth at which the Chl profile returns to a background value, computed empirically from Z_m and Z_{eu} , the euphotic depth)

Day of the year of the satellite-Argo matchup

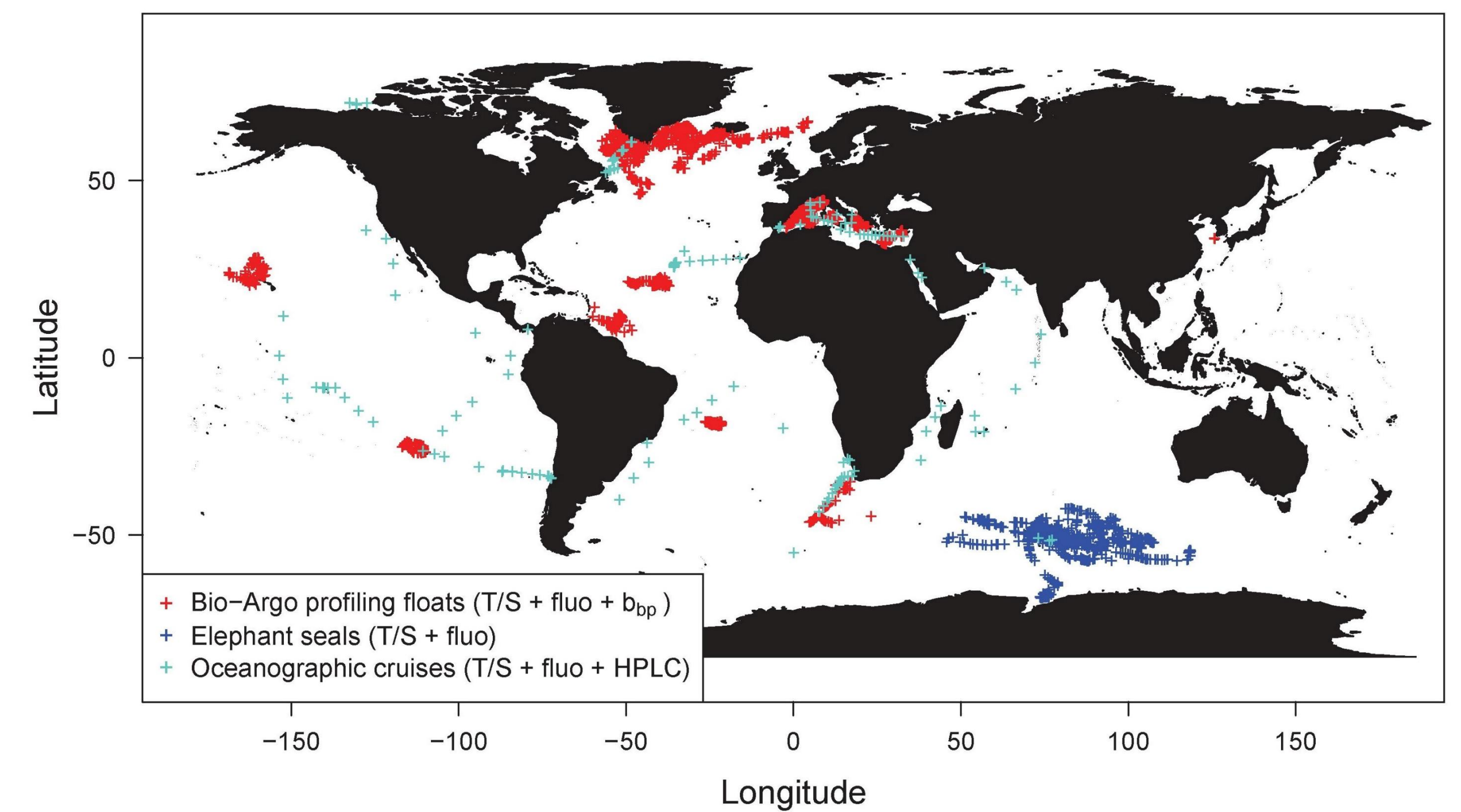
Input layer

Hidden layers

Output layer

Vertical distribution of the bio-optical properties

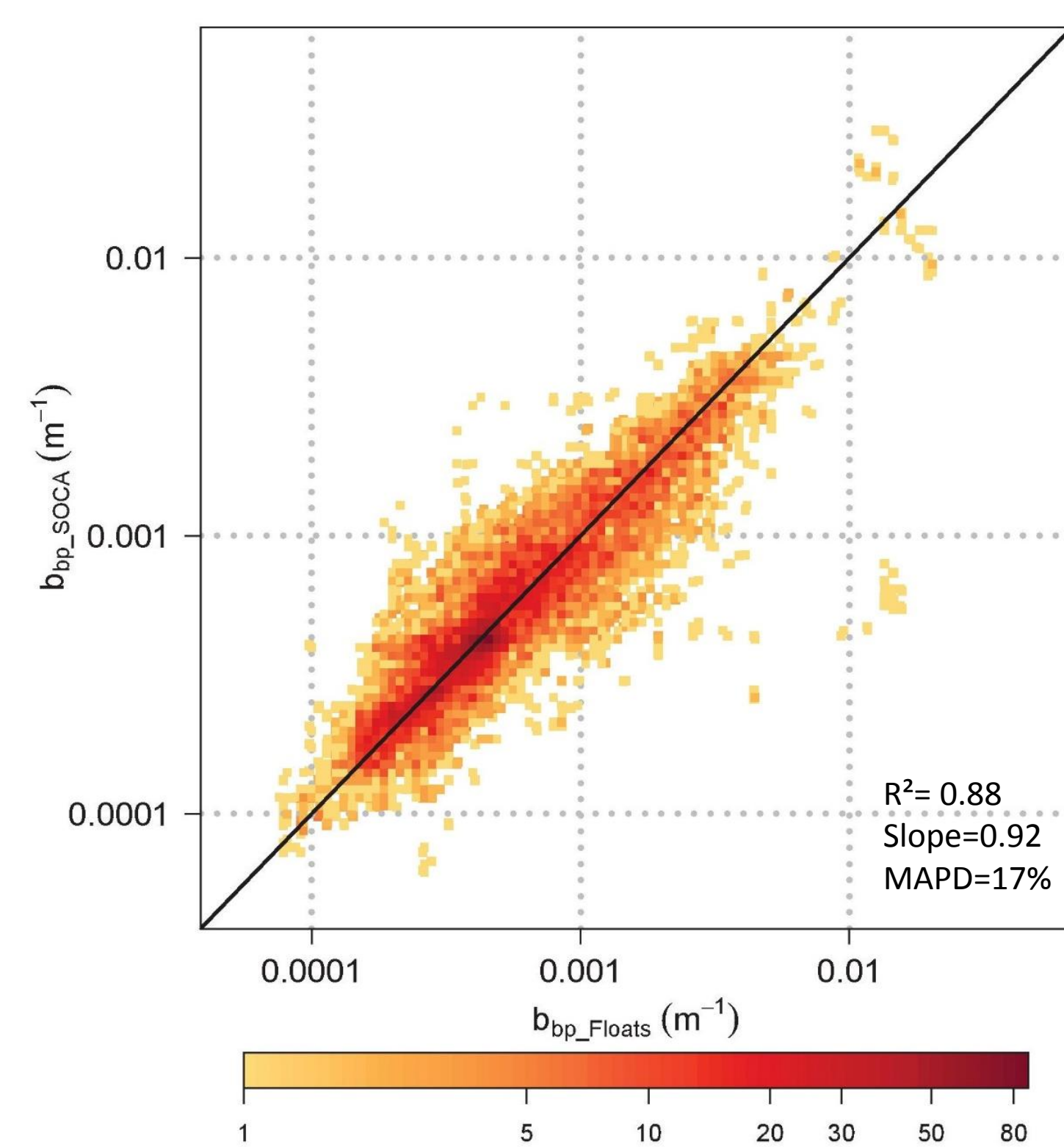
- For b_{bp} : 10 simultaneously retrieved values of b_{bp} at 700nm for the water column
- For Chl: 10 normalized fluorescence values along the water column. Then the FLAVOR method (Sauzède et al., 2015) is applied to retrieve the [Chl] and phytoplankton community composition



Data used for the training and validation of the MLPs: 1) 3,190 **Bio-Argo** profiles of $b_{bp}(700nm)$ and remotely-sensed values of b_{bp} for SOCA-BBP; 2) 4,902 profiles of fluorescence (from **Bio-Argo floats**, **elephant seals** equipped with a fluorimeter and **oceanographic cruises**) and remotely sensed values of [Chl] for SOCA-CHL. This dataset is **representative of the global ocean** in terms of physical and biogeochemical conditions.

b_{bp} retrieval

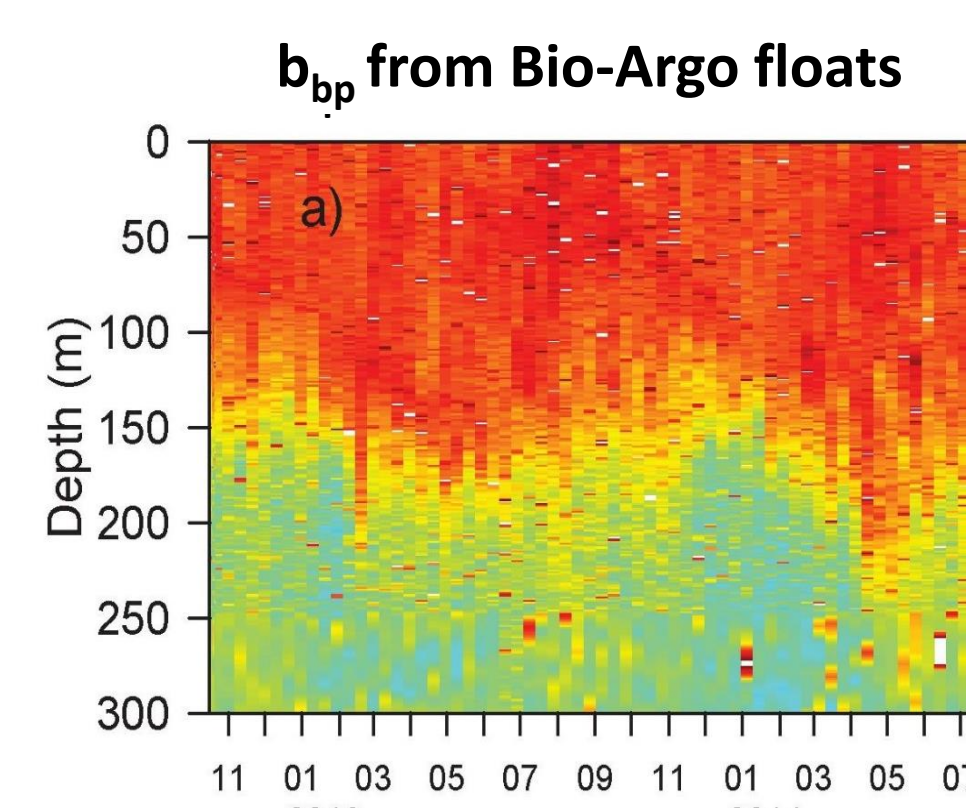
Independent validation using 20% of the initial database (907 profiles)



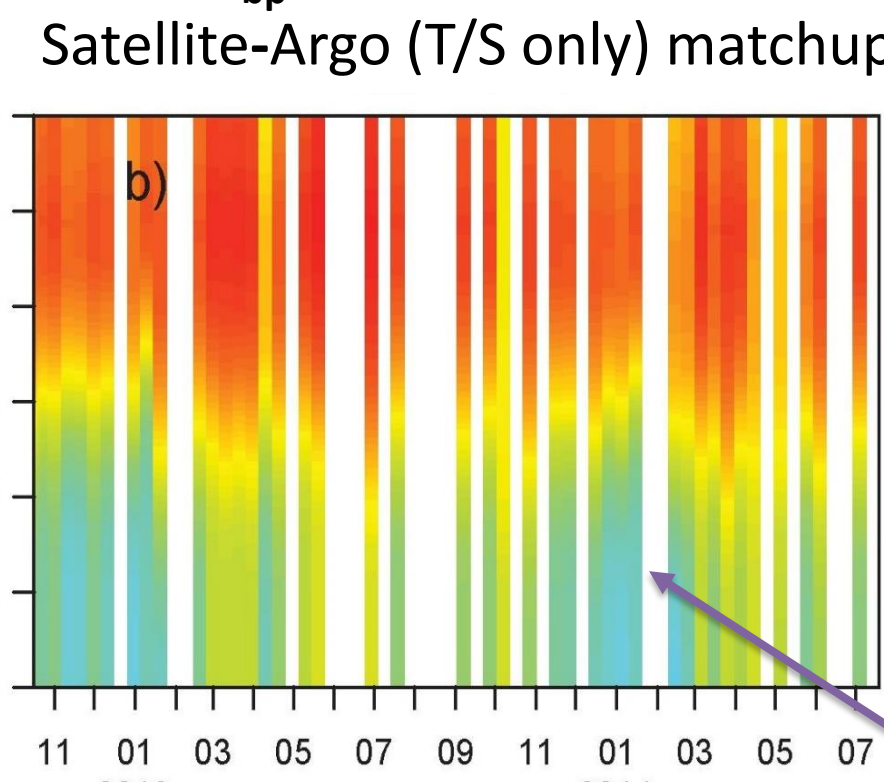
Comparison of b_{bp} retrieved by SOCA-BBP and reference values measured by the Bio-Argo profiling floats. The color bar indicates the number of values in each pixel. MAPD: Median Absolute Percent Difference

Validation using time series of "independent" floats (not integrated in the training and validation databases)

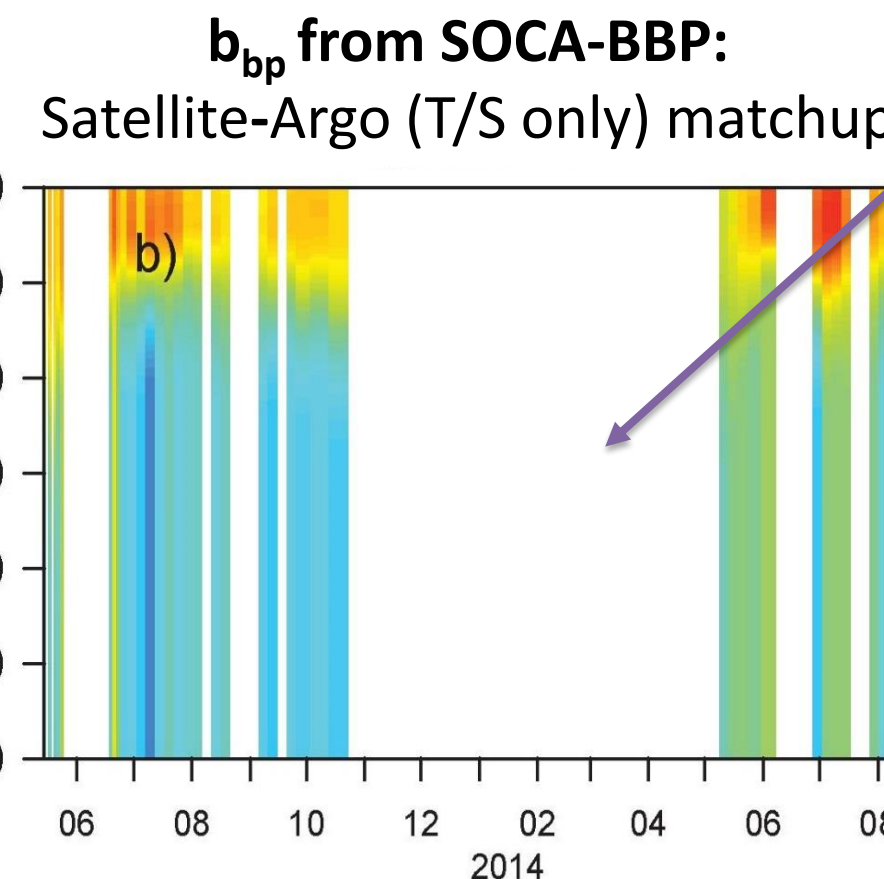
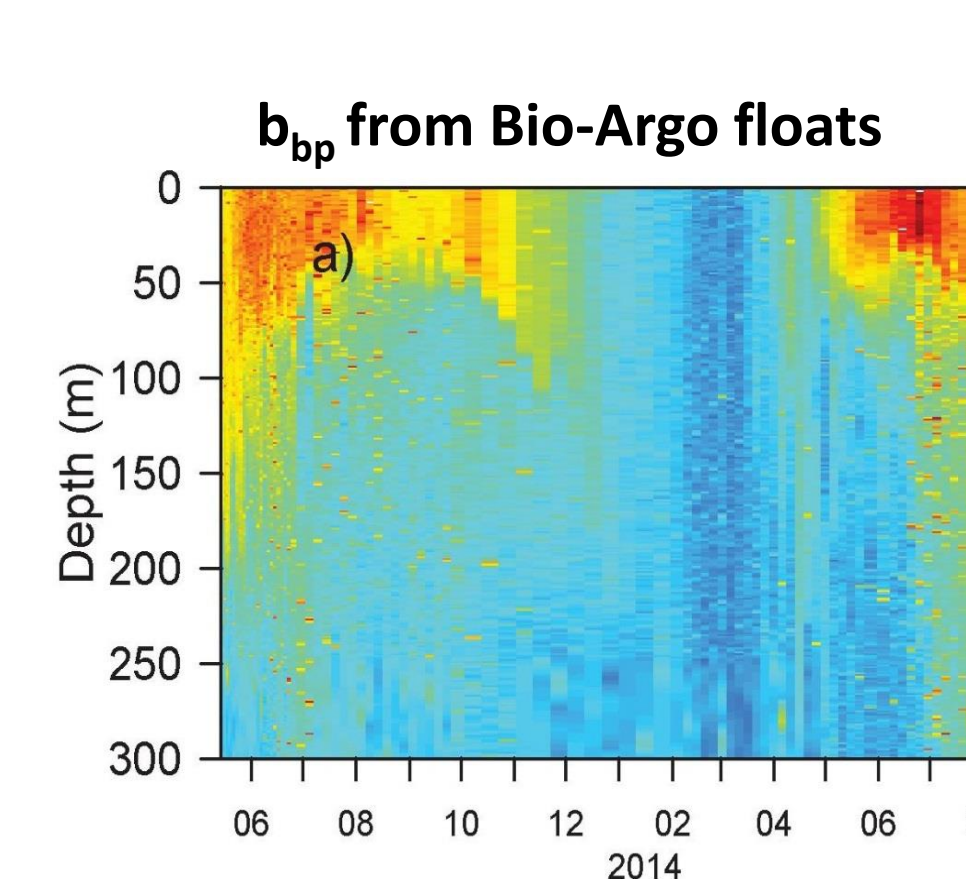
North Atlantic Subtropical Gyre



b_{bp} from SOCA-BBP:



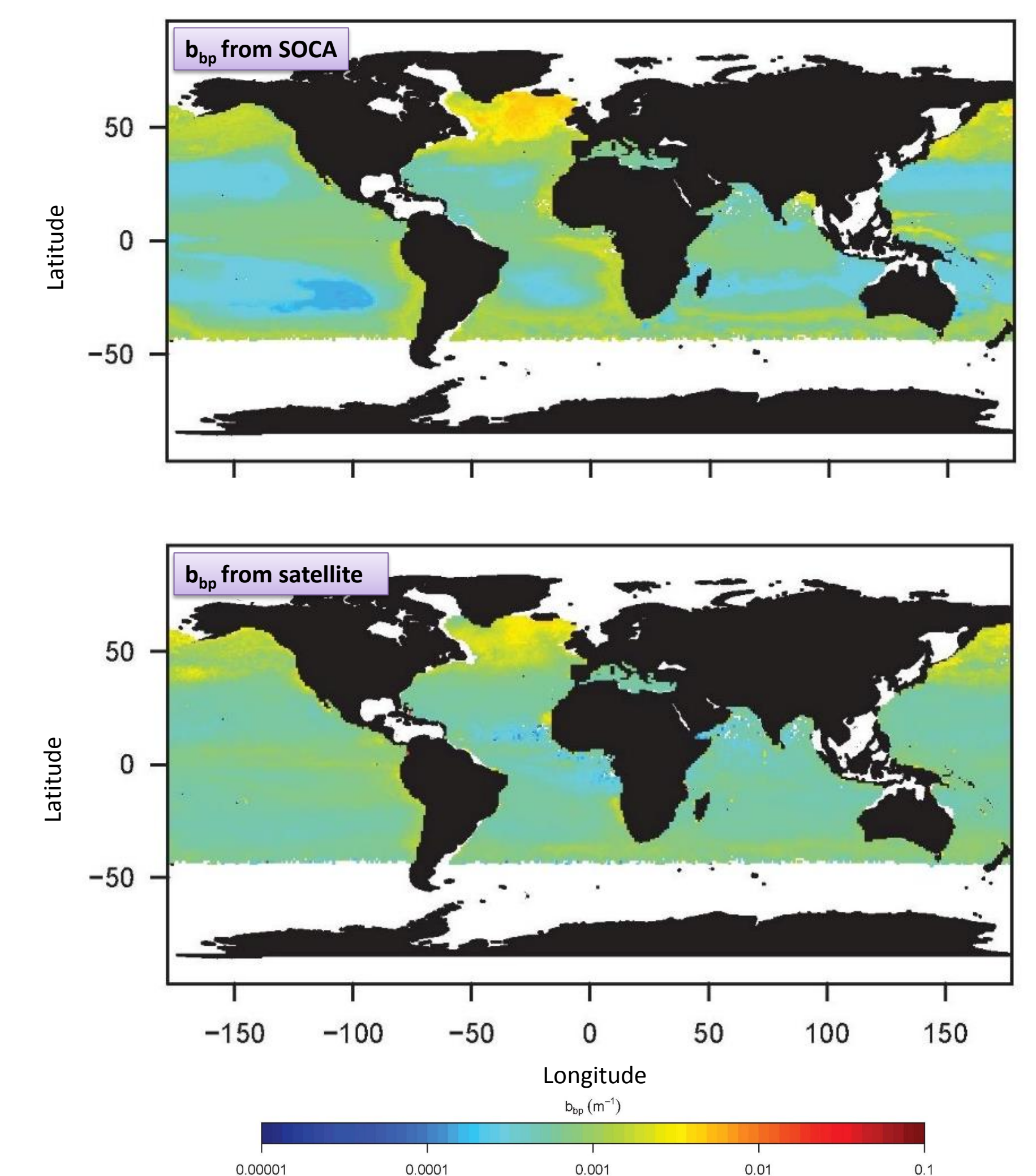
North Atlantic (Irminger and Labrador Sea)



Comparison of SOCA-retrieved vertical profiles and reference Bio-Argo profiles of b_{bp}

Global application of SOCA-BBP

Satellite climatology + Argo T/S climatology (Roemmich and Gilson, 2009)



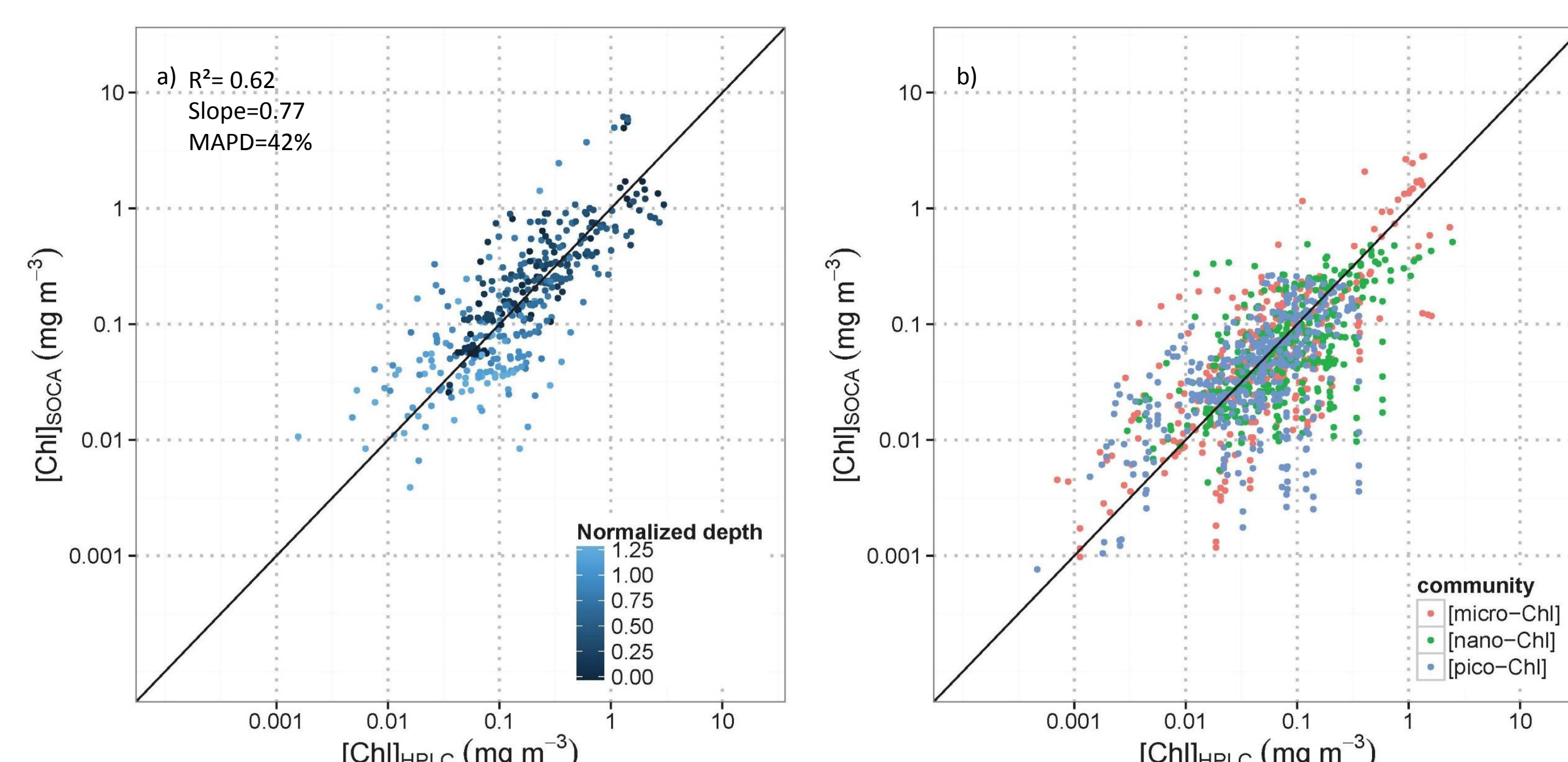
Comparison of SOCA-retrieved global climatology and climatological satellite estimates of b_{bp} for the month of June

→ The first results of the application of SOCA-BBP at the regional and global scales are very promising

Chl retrieval

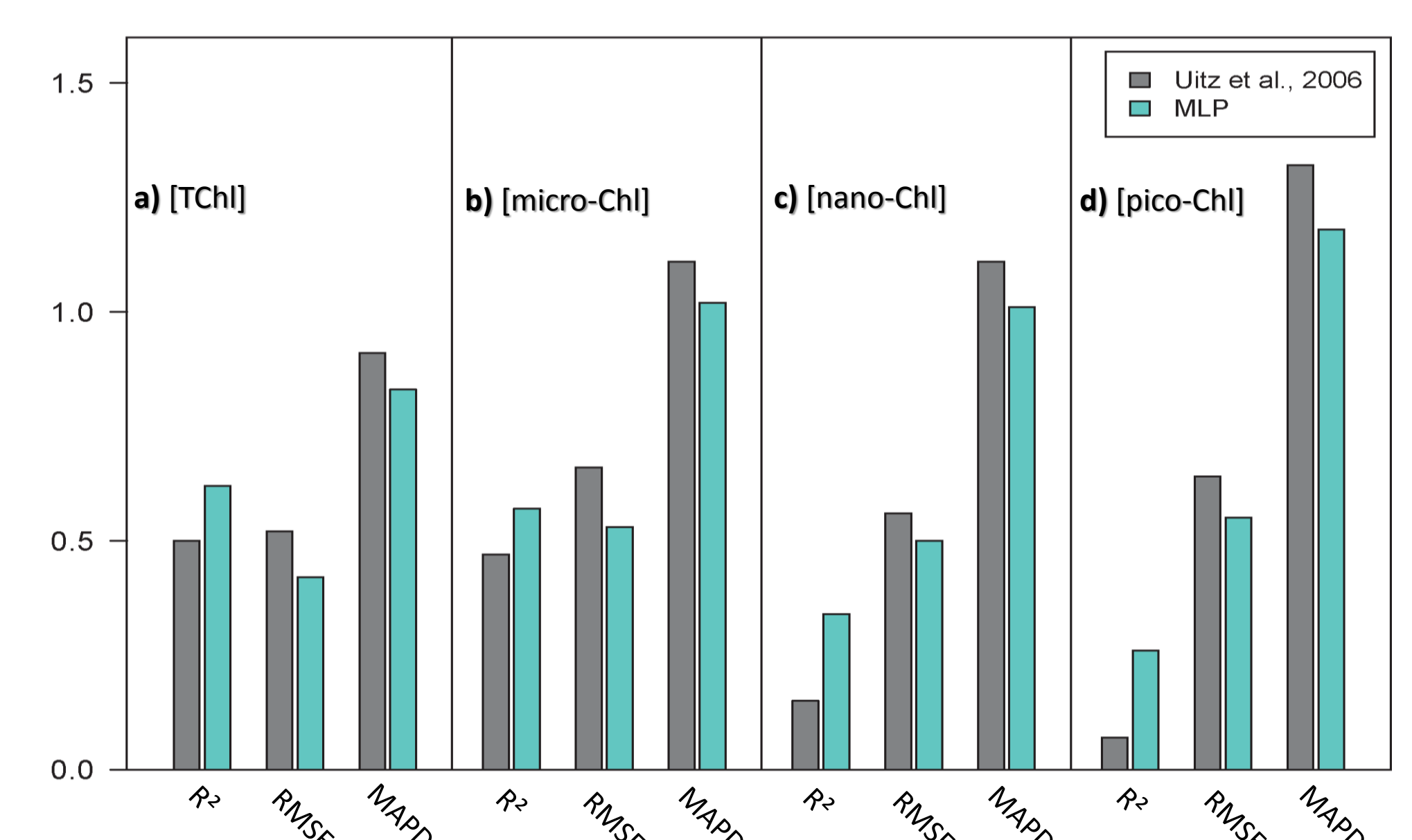
Validation with HPLC data

SOCA-CHL is validated using 41 pigment profiles determined by HPLC (High Performance Liquid Chromatography) and collected during 7 oceanographic cruises.



Comparison of [Chl] retrieved by SOCA-CHL and the HPLC reference values for: a) the total phytoplankton biomass, [Chl], according to the normalized depth of retrieval and b) the phytoplankton community composition.

Comparison with the model of Uitz et al. (2006)



Comparison between [Chl] retrieved by SOCA-CHL and by the model of Uitz et al. (2006). RMSE= Root Mean Squared Error MAPD= Median Absolute Percent Difference

→ SOCA-CHL improves the retrieval of the [Chl] associated with the total phytoplankton biomass and with the three size classes compared to the model of Uitz et al. (2006)